

Frontend Assignment — Live Stock Watch List

React Native take-home assignment

Brief Objective

Build a frontend application that displays a very large list of stocks and keeps their prices updated in real time by polling the provided API.

The goal of this assignment is not only correctness, but also to demonstrate your ability to build performant UIs when working with large datasets.

Provided Resources

1. scrips.json

A JSON file containing approximately 800 stocks. Each stock contains the information required to render the list (such as symbol and name).

2. Price API

You can fetch the latest market data for one or more stocks using the following endpoint.

Endpoint

```
POST https://api.v2.liquide.life/api/markets/ohlcv
```

Headers

```
Content-Type: application/json
```

Example Request

```
curl --location 'https://api.v2.liquide.life/api/markets/ohlcv' \
--header 'Content-Type: application/json' \
--data '{
  "symbols": ["SYMBOL1", "SYMBOL2"]
}'
```

Example Response

```
{
  "status": "success",
  "code": 200,
  "count": 2,
  "message": {
    "data": [
      {
        "symbol": "SYMBOL1",
        "isin": "INE00C30xxxx",
        "ltp": 127.2,
        "open": 130.61,
        "prevClose": 130.5,
        "high": 131.07,
        "low": 126.8,
        "volume": 543574,
        "timeStamp": "2026-07-23T15:30:00",
        "scripId": "",
        "navDate": null,
        "tsn": 1114
      },
      {
        "symbol": "SYMBOL2",
        "isin": "ISINxxx",
        "ltp": 23869.6,
        "open": 23904.8,
        "prevClose": 23996.25,
        "high": 23990.75,
        "low": 23807.2,
        "volume": 0,
        "timeStamp": "2026-07-23T15:31:03",
        "scripId": "",
        "navDate": null,
        "tsn": 89994
      }
    ]
  }
}
```

Requirements

Build a page that:

- Displays all stocks from scrips.json in a scrollable list.
- Continuously refreshes prices by polling the API.
- Updates prices in the UI as new data is received.
- Frontend specific requirements:
 - Add a gradient (any color combination of your choice) to each line item in the list.
 - A maximum of 6–7 line items must fit on the screen at any given scroll position.
 - Each line item must contain: Stock symbol, the realtime LTP (from polling API), as well as realtime percentage change (can be calculated simply from current LTP and prevClose value from the API).

Notes

- The API accepts an array of stock symbols and returns the latest market data for those symbols.
- Polling frequency is left to your discretion.
- Feel free to use any React libraries or utilities that you think are appropriate.
- Some basic layout instructions are mentioned in the requirements section. The exact design and UI is left to your free will — a neat UI would be good.

Evaluation Criteria

We will evaluate your submission on:

Correctness Code Quality React Best Practices **Performance - Top Priority** User Experience

What We're Looking For

The dataset intentionally contains about 800 stocks. Consider how your implementation behaves as the user scrolls through the list and how it scales as the number of items grows. **We're looking for how efficiently your solution can manage the re-rendering of hundreds of items in the list.**

We're particularly interested in the efficiency of your solution rather than simply making it work.

Submission

Please share:

- Source code
- Instructions to run the project locally
- A short note describing any assumptions/trade-offs/enhancements made during implementation

Note:

- Prices only update live during market hours (Mon–Fri, 9:15 AM to 3:30 PM).
- Consider testing your solution during at least one market session.
- The walkthrough in the next interview round will also be conducted during market hours to put the solution to the test.